

Multi-Agent AI Systems for Stock Analysis: An Agentic Approach to Recommendation and Explanation

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Abstract

This paper presents a multi-agent large language model (LLM) framework designed to both explain analytical recommendations and recommend equity stocks using structured financial data, technical indicators, and real-world news. The system consists of two coordinated modes: an Explainer agent that produces transparent, data-grounded reasoning to interpret human analyst ratings, and a Recommender agent that generates independent buy/sell/hold decisions through a hierarchical synthesis of Fundamental, Technical, and News specialist agents. A human study evaluates the plausibility and consistency of the Explainer’s outputs, achieving strong scores across key analytical dimensions. Preliminary evaluation of the Recommender agent on a limited sample ($n=100-300$) showed insufficient statistical power to draw reliable conclusions about predictive performance. This limitation motivated our development of a comprehensive evaluation framework that addresses scalability, realistic simulation, and risk-adjusted metrics—providing a roadmap for rigorous validation of agentic investment systems against actual market outcomes.

Introduction

Motivation

Financial markets integrate an increasingly dense ecosystem of quantitative indicators, qualitative signals, and narrative-driven sentiment. In addition to traditional fundamentals such as earnings, leverage, and cash flows, investors must now process high-frequency market data, macroeconomic updates, regulatory filings, supply-chain disruptions, and news-driven shifts in investor psychology. This informational explosion has transformed equity analysis into a cognitively intensive task requiring multi-modal reasoning over heterogeneous evidence.

Conventional prediction models—particularly those relying on purely numerical, time-series, or factor-based representations—struggle to incorporate unstructured information such as corporate disclosures, text-based analyst reports, and real-time news sentiment. Meanwhile, human analysts face limits in breadth and speed: report coverage varies across firms, attention is unevenly distributed, and the manual synthesis of cross-source evidence increasingly fails to scale in modern markets (Kuruvilla and Moorthi 2025).

As a result, both statistical models and human analysts encounter persistent blind spots in identifying connections between firm events, performance dynamics, and market reactions.

The Emergence of Agentic AI in Finance

Large Language Models (LLMs) have introduced new possibilities for processing unstructured financial information, from summarizing earnings calls to extracting sentiment from news and filings. However, traditional single-pass LLM systems are limited to static input-output behavior. They lack the ability to autonomously retrieve relevant evidence, verify claims, plan multi-step analysis, or integrate structured numerical data with unstructured documents.

Agentic AI systems address these limitations. These architectures embed LLMs within a scaffold of retrieval modules, planning algorithms, memory stores, and tool execution capabilities, enabling multi-step reasoning and autonomous decision workflows. Agentic frameworks such as AutoGen, CrewAI, LangGraph, and other emerging orchestration systems demonstrate that LLMs can iteratively refine hypotheses, call external tools, and decompose complex analytic tasks (Maragheh and Deldjoo 2025). In finance, early applications include portfolio rebalancing agents, automated compliance systems, and real-time risk monitors. These systems show strong potential to coordinate information across structured databases and unstructured text—but current work remains narrow, focusing on isolated tasks rather than end-to-end analyst workflows.

Critically, no existing research offers an agentic architecture that replicates the holistic reasoning pipeline of human equity analysts. Prior systems typically perform one of the following in isolation:

- Text-driven sentiment analysis
- Numerical factor extraction
- Event classification
- LLM-based query answering

What is missing is a unified, agentic workflow that autonomously retrieves firm fundamentals, links events to financial outcomes, integrates narrative information from news, and generates structured investment rationales alongside independent buy/sell/hold decisions. As a result, analysts lack an AI system capable of both interpreting histori-

cal recommendations and producing its own evidence-based judgments.

Contributions

An Agentic Framework for Equity Recommendations

This paper introduces the Agentic Stock Recommendation System, a multi-agent LLM-driven framework designed to emulate and extend traditional equity analysis. Built using the CrewAI orchestration framework, the system integrates structured financial datasets with unstructured narrative and event information to support autonomous, evidence-grounded investment reasoning. Our system constructs a standardized fact pack by linking:

IBES	Analyst recommendations
Compustat	Firm fundamentals
CRSP	Returns and market characteristics
Capital IQ Key Developments	Detailing firm-specific events

Table 1: Data Source Integration

Leveraging this evidence, the system deploys two complementary agentic roles:

- **Explainer Agent:** Generates natural-language rationales that interpret and contextualize historical analyst decisions.
- **Recommender Agent:** Produces independent buy/sell/hold recommendations grounded entirely in retrieved evidence, without referencing prior analyst outputs.

Our key contributions are as follows:

1. **An end-to-end agentic architecture for equity analysis.** We present the first system that unifies explanation generation and independent stock recommendations within a multi-agent LLM-driven workflow.
2. **A fact-pack retrieval and synthesis pipeline for financial reasoning.** We propose a novel retrieval architecture that integrates structured financial databases with unstructured firm event and news information, enabling multi-modal agentic reasoning.
3. **A scalable, reproducible framework for analyst-style financial decision modeling.** Our approach offers a standardized methodology for evaluating whether agentic AI can emulate human financial reasoning, reduce information overload, and generate coherent, evidence-grounded investment insights.
4. **Bridging structured fundamentals, narrative intelligence, and agentic reasoning.** This work aims to advance the role of agentic AI in financial analysis and create new pathways for scalable, transparent, and interpretable decision support in equity markets.

Background

Traditional Analysis and Non-Agentic Tools

Equity analysis has historically relied on manual workflows where analysts synthesize firm fundamentals, macroeco-

nommic indicators, and qualitative narratives from earnings calls and reports. This process is cognitively intensive and difficult to scale (Shah et al. 2025). While computational tools like screening filters and factor-based models aid in data aggregation, they serve primarily as static decision support. They lack the reasoning capability to autonomously link disparate pieces of evidence—such as a sudden CEO departure and a dip in cash flow—into a coherent investment thesis. These limitations create a need for systems capable of autonomous, multi-modal reasoning.

Machine Learning and the Interpretability Gap

To address scalability, the industry has adopted machine learning (ML) models, including random forests and deep learning architectures (Bhattacharjee and Bhattacharja 2019; Zohren and Roberts 2024). While effective at capturing temporal dynamics in numerical data, these systems face critical barriers in professional equity analysis:

- **Black-Box Nature:** ML models rarely offer auditable justifications for their predictions, a requirement for regulatory compliance and investor trust.
- **Data Modality Limits:** Standard ML pipelines struggle to incorporate unstructured text (news, filings), missing the qualitative context often driving market moves (Kuruvilla and Moorthi 2025).
- **Lack of Reasoning:** ML systems function as static prediction engines, unable to perform the multi-step planning or hypothesis testing characteristic of human analysts.

LLMs for Financial Text Processing

Large Language Models (LLMs) like FinBERT (Liu et al. 2021) and BloombergGPT (Wu et al. 2023) have bridged the gap in processing unstructured text, outperforming general models in sentiment classification and event extraction. However, current LLM applications in finance remain largely "single-step" processes. They map text to sentiment scores or summaries but lack the architecture to autonomously retrieve verification data, compare current events to historical patterns, or plan complex analytical workflows. Consequently, they cannot independently generate the holistic, evidence-grounded rationales found in sell-side reports.

The Gap: Agentic Frameworks in Finance

Agentic AI represents a paradigm shift where LLMs are embedded within architectures capable of planning, tool execution, and memory management (Maragheh and Deldjoo 2025). Early financial agents have successfully tackled narrow tasks such as algorithmic trading execution, risk alert escalation, and automated compliance checking.

However, a significant gap remains: there is no existing agentic framework designed to emulate the end-to-end workflow of an equity analyst. Current systems do not bridge the divide between structured financial databases (e.g., Compustat) and unstructured narratives, nor do they produce independent, explainable buy/sell recommendations. This paper addresses that gap by proposing a unified architecture

that combines the quantitative rigor of data retrieval with the narrative explanatory power of LLM-driven agents.

Methodology

Data Preprocessing

We construct our dataset by integrating four major sources accessed through Wharton Research Data Services (WRDS): Compustat for quarterly fundamentals, CRSP for daily prices and volume, IBES for analyst recommendations, and Capital IQ for recent news and key developments. Data points were collected from 1-Jan-2008 to 31-Dec-2024. After merging these datasets using CUSIP identifiers, we align all variables at the daily level to ensure consistent coverage across sources. Fundamental variables are forward-filled to handle missing observations and maintain continuity, while technical indicators are computed directly from CRSP to capture market dynamics. News headlines from Capital IQ are cleaned, deduplicated, and restricted to a ± 30 -day window for the Explainer (centered on recommendation date) and a 30-day historical window for the Recommender (only days before recommendation). This robust preprocessing pipeline produces a high-quality, well-aligned dataset that our agent uses to perform downstream equity analysis and reasoning.

Agentic Architecture Overview

The Multi-Agent Analyst system is a sophisticated AI architecture that uses specialized agents to analyze stock recommendations. The system operates in two distinct modes: Explainer Mode, which explains why a human analyst gave a specific rating, and Recommender Mode, which generates an independent AI-driven rating recommendation. Both systems follow a hierarchical multi-agent architecture with:

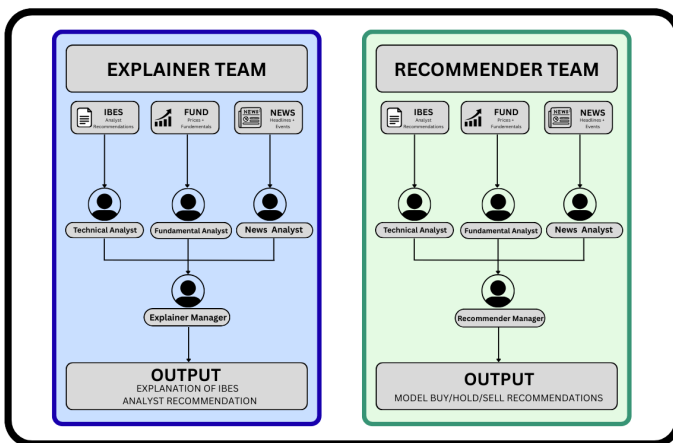


Figure 1: Backend Overview of Agentic Framework

1. **3 Specialist Analysts:** Fundamental, Technical, and News analysts.
2. **1 Manager/Synthesis Agent:** Synthesizes all specialist reports into the final output.

3. **Sequential Execution:** Specialists run first, then the manager synthesizes.

The Explainer Agent

The Explainer Team is designed to clarify why a human equity analyst assigned a particular rating to a stock. To accomplish this, the system uses four specialized AI agents that collaborate to interpret three primary data modalities—fundamental, technical, and news—before synthesizing their findings into a coherent explanation. The process begins with an input drawn from the IBES dataset, providing the historical recommendation details (ticker, date, rating, analyst name). For example, an analyst might have issued a “SELL” rating for AMZN on 2008-01-08. The system then retrieves all relevant supporting data from WRDS sources to contextualize this recommendation.

After preprocessing, four domain-specific agents are instantiated, each with a defined role, goal, and behavioral constraints. All agents use a low-temperature (0.3) LLM configuration to ensure consistent, evidence-based reasoning:

- **Fundamental Analyst:** Interprets financial statement data, highlights signals, states when metrics are missing, and restricts analysis strictly to fundamentals.
- **Technical Analyst:** Evaluates momentum, trend indicators, volume patterns, and other technical signals without inferring any price levels or metrics beyond what is provided.
- **News Analyst:** Assesses sentiment and event relevance from headlines only, avoiding speculation and classifying news as positive, negative, or neutral.
- **Explainer Manager:** Acts as a senior research director who synthesizes all three specialist reports into a unified explanation of the human analyst’s rating. The manager does not perform new analysis but instead integrates evidence, producing a structured markdown report.

Each agent operates within a structured task specification that dictates input formatting, output requirements, and strict constraints. The system uses a layered prompting strategy incorporating role-based instructions and intelligence weighting to prevent hallucination. Importantly, specialists are not given the human analyst’s rating; only the manager receives it.

Explainer Task and Flow The system runs through a four-stage sequential process:

1. The fundamental, technical, and news crews run independently, each producing a domain-specific markdown report based solely on its respective dataset.
2. The manager crew executes, consuming all three specialist outputs and the IBES metadata to compose the final explanation.

Sequential execution simplifies debugging, avoids rate-limit issues, and ensures the manager has all necessary inputs before synthesizing the final report. The final output is a structured markdown document beginning with an Executive Summary, followed by Key Drivers, an Analyst Per-

Listing 1: Task and Agent Creation for Agentic Workflow

```
def create_[agent]_task(agent, data: str) -> Task:
    description = f"""
    [Detailed instructions for the agent]

    INPUT DATA:
    {data}

    OUTPUT FORMAT:
    [Expected markdown structure]

    CONSTRAINTS:
    [What they cannot do]
    """

    return Task(
        description=description,
        agent=agent,
        expected_output=" [Brief description of expected
        output]",
    )
```

spective section, and a Confidence Score from 0–100 indicating how strongly the evidence supports the human rating. The report concludes with the full fundamental, technical, and news analyses as supporting appendices. Together, this pipeline provides a transparent, modular, and interpretable system for reconstructing the rationale behind historical equity analyst recommendations.

Recommendation Agent

The Recommender Agent uses a similar architecture and data-processing pipeline as the Explainer system. It leverages fundamental, technical, and news-based specialist agents that independently analyze their respective datasets. Each specialist produces a structured, domain-specific assessment using the same task templates and prompting constraints—ensuring consistent, non-hallucinatory, and domain-restricted reasoning.

Once these three reports are generated, they are passed to the Recommender Manager, which functions similarly to the Explainer Manager but with a different objective: instead of explaining a human analyst’s decision, it synthesizes the signals to produce its own Buy/Hold/Sell recommendation. The Recommender Manager evaluates signal alignment across the three specialist reports, assesses the strength and direction of each domain’s findings, and assigns implicit weighting to determine the dominant market signal. Based on this synthesis, the manager outputs a final recommended rating—Buy, Hold, or Sell—along with a brief confidence assessment.

While the backend architecture mirrors the Explainer system, the Recommender’s focus shifts from interpretability to decision-making, producing a concise and actionable investment recommendation rather than a narrative explanation. This dynamic weighting is illustrated in Figure 2.

Prompt Engineering

In our system, prompt engineering plays a central role in shaping the behavior of both the Recommender and the Explainer agents, but in meaningfully different ways.

For the Recommender, prompts enforce a structured, deterministic decision-making process. Each specialist agent received strict instructions to evaluate only its assigned domain, and output a rating, confidence level, and justification using a predefined five-point scale. The Manager prompt synthesizes these outputs with explicit weighting rules, instructions for resolving contradictions, and context-aware guidance tailored to market regimes.

For the Explainer, prompt engineering focuses on constraining the model’s reasoning rather than its decision-making. The Explainer’s role is to interpret the final recommendation rather than generate one, so its prompts emphasized reasoning exclusively from the provided data, avoiding any external knowledge or hallucinated logic.

This distinction, decision-generation versus explanation-generation, required separate prompt strategies: deterministic, rating-oriented constraints for the Recommender, and evidence-bounded, chain-of-thought-restricted instructions for the Explainer. Together, these prompt engineering techniques ensured that each component of the system behaved reliably, transparently, and strictly within its intended scope.

View Listing 2 at the end of the document for direct examples of prompt engineering within our Agentic workflow.

Table 2: Comparison of Explainer vs Recommender

Aspect	Explainer	Recommender
Goal	Explain human’s rating	Make own rating
Specialist Output	Analyze report	Rating + report
Manager Output	Explanation synthesis	Rating decision + weighting
Shows Human Rating?	Yes (explaining it)	No (hidden until after)
Confidence	Explanation conf. (0–100)	Rating conf. (High-/Med/Low)
Temperature	0.3 (more consistent)	0.2 (more consistent)
Key Innovation	Multi-perspective explanation	Intelligent rating synthesis
Data Window	FUND: 90 days, NEWS: ± 30 days	FUND: latest available, NEWS: 30 days before
LLM Instance	Per-agent instances	Shared instance

Evaluation

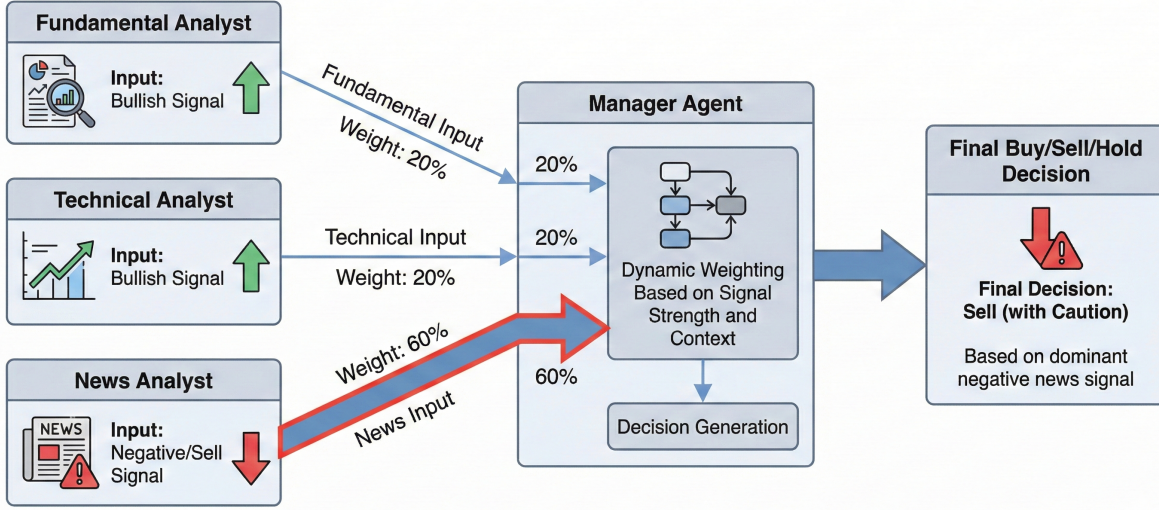
Explainer Agent Evaluation

To evaluate the Explainer agent, we conducted a small human study in which participants assessed the quality of 16 high-quality samples selected based on data completeness (stocks with no missing data). Each explanation was evaluated across three dimensions:

1. **Plausibility:** Whether the reasoning resembled that of a real financial analyst.

Dynamic Weighting in Multi-Agent LLM Recommender System

Scenario: Negative News Overrides Positive Fundamentals and Technicals



Example: Negative news dominates positive fundamentals and technicals due to market sensitivity to scandals.

Figure 2: Weighting Scenario for Recommender Agent

2. **Signal Coverage:** Measuring whether the explanation incorporated all three input signals (fundamental, technical, and news).
3. **Internal Consistency:** Assessing whether the final conclusion logically followed from the evidence presented.

Each dimension was rated on a 1–5 scale, and the results—summarized in Table 3—show that all three categories achieved average scores of at least 4.5. These findings suggest that the Explainer agent reliably produces coherent, data-grounded, and realistic analytical justifications. Additionally, the agent provides a confidence score at the end of each explanation. This score reflects the relative strength and clarity of the underlying data sources as well as the weights implicitly assigned to them during the reasoning process. Explanations grounded in strong, consistent signals receive higher confidence ratings, while those relying on weaker or conflicting evidence are assigned lower confidence.

Table 3: Average Human Evaluation Scores for Explainer Agent (1–5 Scale)

Factor	Average Score
Plausibility	4.81 / 5.0
Signal Coverage	4.50 / 5.0
Internal Consistency	4.50 / 5.0

Recommender Agent Evaluation

Overview and Methodology The Recommender agent generates independent investment signals that must be assessed against market outcomes. We utilize a backtesting

framework that simulates trading strategies based on the model’s outputs. The pipeline involves:

- **Signal Generation:** Model ratings (e.g., Strong Buy) are mapped to trading signals (Long/Short/Neutral).
- **Return Computation:** Forward-looking returns are calculated at 1-month and 3-month horizons.
- **Performance Metrics:** We track Directional Accuracy and Average P&L per trade, benchmarked against IBES analyst ratings and an “Always-Long” baseline.

Limitations and Preliminary Results Initial evaluation encountered significant **sample size constraints** due to the computational cost of the multi-agent pipeline (sequential LLM calls per recommendation). With only 100–300 historical recommendations evaluated, the sample was insufficient for statistically significant conclusions. Furthermore, the framework uses simplified execution assumptions, ignoring real-world factors like transaction costs. While feasibility is demonstrated, definitive claims regarding investment performance require larger-scale testing.

Framework for Scaled Backtesting To enable rigorous validation, future work requires significant enhancement of the evaluation framework:

- **Scalability:** Implement parallel batch processing to achieve statistical power with 5,000+ historical recommendations.
- **Realistic Simulation:** Incorporate transaction costs and risk-adjusted position sizing based on the Portfolio Manager’s confidence scores.
- **Advanced Metrics & Ablation:** Employ risk-adjusted metrics (Sharpe/Sortino ratios) and conduct ablation stud-

ies to isolate the predictive contribution of the specialist agents.

This roadmap focuses on transforming the Recommender into a verifiable quantitative research tool, bridging AI interpretability with financial performance.

Conclusion and Future Work

This work demonstrates that an agent-based LLM framework can generate both coherent explanations and actionable investment recommendations using heterogeneous financial signals. The Explainer agent produced highly plausible, well-structured, and data-grounded justifications, achieving strong human-evaluation scores across plausibility, signal coverage, and internal consistency. Its integrated confidence scoring mechanism provides an interpretable measure of how strongly each signal contributed to the final explanation. Meanwhile, the Recommender agent showed that alignment with human analysts is not necessarily the ideal benchmark—its significant divergence from IBES ratings motivated a more meaningful, performance-based evaluation. By directly comparing model-generated and human-generated signals against actual market outcomes, this work repositions LLMs not as imitators of analyst opinion but as potential independent decision-makers. Together, these findings highlight that LLM-driven financial agents can operate with transparency, consistency, and a level of reasoning that is both structured and human-auditable.

Limitations and Next Steps

The stakes of this project extend beyond academic interest. As LLMs become increasingly integrated into financial decision-making pipelines, the ability to generate interpretable, auditable, and data-grounded outputs is essential—not only for regulatory compliance, but also for building trust in automated investment systems. Understanding when and why a model reaches a conclusion, and whether that conclusion would have been profitable historically, has direct implications for asset managers, risk officers, and retail investors alike. In this sense, our agentic framework provides a tangible step toward AI systems that can justify their recommendations with domain-relevant reasoning rather than opaque intuition.

1. **Expand the Human Study:** Increase the number and diversity of participants to evaluate additional dimensions such as risk identification, company-specific understanding, and analytical vs. emotional tone.
2. **Improve the Recommender Benchmark:** Scale backtesting to a larger set of model-generated signals to reduce noise and better assess true profitability.
3. **Adopt Domain-Specific Financial Models:** Integrate models such as FinGPT, FinBERT, or BloombergGPT to decrease hallucinations and improve financial reasoning precision.
4. **Integrate MCP Server Capabilities:** Use Model Context Protocol (MCP) servers to enable dynamic interaction with financial APIs, data terminals, retrieval systems,

and portfolio simulators—allowing agents to query, update, and validate information in real time.

In sum, this project provides an early blueprint for the future of LLM-powered financial analysis—one where models serve not only as passive text generators but as transparent, modular, and evaluable agents capable of assisting in real investment workflows.

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Listing 2: Examples of Prompt Engineering

```
def create_recommender_manager_task(
    fundamental_report: str,
    technical_report: str,
    news_report: str,
    stock_info: str = ""
) -> Task:

    description = f"""
    You are the Portfolio Manager synthesizing three
    analyst
    reports into a final investment recommendation.

    {stock_info}

    FUNDAMENTAL REPORT: {fundamental_report}
    TECHNICAL REPORT: {technical_report}
    NEWS REPORT: {news_report}

    Your job:
    1. Extract rating (StrongBuy...Sell)
    2. Assess confidence level
    3. Identify signal alignment
    4. Apply weighting logic based on:
       - Signal strength & Data quality
       - Market context
    5. Make final decision: ONE rating
    6. Explain weighting logic clearly

    OUTPUT FORMAT (strict):
    ## Model Recommendation (Final)
    ### Final Rating
    - **Model Rating**: <Buy/Hold/Sell...>
    - **Overall Confidence**: <High/Med/Low>

    ### Analyst Ratings Summary
    - **Fundamental**: <rating> (conf: <level>)
    - **Technical**: <rating> (conf: <level>)
    - **News**: <rating> (conf: <level>)

    ### Signal Alignment
    - **Agreement**: <Agree/Split/Disagree>
    - **Dominant Signal**: <Which view wins?>

    ### Weighting Logic
    <Explain how you weighted each input>

    ### Synthesis
    <3-4 sentences explaining final decision>

    ### Risk Assessment
    - **Primary Risk**: <biggest downside>
    - **Data Quality**: <missing data?>

    CONSTRAINTS:
    - Start with '## Model Recommendation (Final)'
    - Choose ONE clear final rating
    """

    return Task(
        description=description,
        agent=recommender_manager,
        expected_output="Markdown starting with '## Model Recommendation'"
    )
```